Solutions - Part 1

1. Introduction and Types of Solutions

A **solution** is defined as a homogeneous mixture of two or more components. Being "homogeneous" means that the composition and properties are uniform throughout the mixture.

- **Solvent:** The component present in the largest quantity. It determines the physical state of the solution.
- **Solute:** The component(s) present in smaller quantities.
- **Binary Solution:** A solution consisting of exactly two components (one solute and one solvent).

Types of Solutions: Solutions can exist as gases, liquids, or solids depending on the solvent.

- **Gaseous Solutions:** Solvent is a gas (e.g., Oxygen and Nitrogen mixture, Chloroform in Nitrogen gas).
- **Liquid Solutions:** Solvent is a liquid (e.g., Ethanol in water, Oxygen dissolved in water, Glucose in water).
- **Solid Solutions:** Solvent is a solid (e.g., Copper in gold, Solution of hydrogen in palladium).

---

2. Expressing Concentration of Solutions

Concentration describes the exact amount of solute dissolved in a specific amount of solvent or solution.

A. Mass and Volume Based Units

- **Mass Percentage (w/w):** The mass of the component per 100 grams of the total solution.
  $\text{Mass } \% = \left(\frac{\text{Mass of component}}{\text{Total mass of solution}}\right) \times 100$
- **Volume Percentage (V/V):** The volume of the component per 100 parts volume of the solution. Commonly used for liquid-liquid solutions (e.g., antifreeze in cars).
  $\text{Volume } \% = \left(\frac{\text{Volume of component}}{\text{Total volume of solution}}\right) \times 100$

- **Mass by Volume Percentage (w/V):** Mass of solute (in grams) dissolved in 100 mL of the solution. Widely used in medicine and pharmacy.
- **Parts per Million (ppm):** Used when a solute is present in very small (trace) quantities.
  $\text{ppm} = \left(\frac{\text{Number of parts of the component}}{\text{Total number of parts of all components}}\right) \times 10^6$

B. Mole Based Units

- **Mole Fraction (x):** The ratio of the number of moles of one component to the total number of moles in the solution.
  For a binary mixture of A and B:
  $x_A = \frac{n_A}{n_A+n_B}$
  Note: The sum of all mole fractions in a solution is always equal to 1 ($x_1 + x_2 + ... + x_i = 1$).
- **Molarity (M):** The number of moles of solute dissolved in 1 Litre (or 1 cubic decimetre) of the solution.
  $M = \frac{\text{Moles of solute}}{\text{Volume of solution in Litres}}$
- **Molality (m):** The number of moles of solute dissolved per kilogram (kg) of the solvent.
  $m = \frac{\text{Moles of solute}}{\text{Mass of solvent in kg}}$

Temperature Dependence: Mass %, ppm, mole fraction, and molality are **independent** of temperature because they rely solely on mass. Molarity changes with temperature because the volume of liquids expands or contracts with heat.

---

3. Solubility

Solubility is the maximum amount of a substance that can be dissolved in a specified amount of solvent at a specific temperature and pressure. It follows the rule **"Like dissolves like"**—polar solutes dissolve in polar solvents (e.g., salt in water), and non-polar solutes dissolve in non-polar solvents (e.g., naphthalene in benzene).

A. Solubility of a Solid in a Liquid

- **Dissolution vs. Crystallisation:** When a solid solute is added to a solvent, it dissolves (dissolution). Simultaneously, dissolved solute particles collide with solid particles and separate out (crystallisation).
- **Dynamic Equilibrium:** The point where the rate of dissolution equals the rate of crystallisation ($\text{Solute} + \text{Solvent} \rightleftharpoons \text{Solution}$).

- **Saturated Solution:** A solution in dynamic equilibrium where no more solute can be dissolved at that temperature and pressure.
- **Effect of Temperature:** Follows Le Chatelier’ s Principle.
  - If dissolution absorbs heat (*endothermic*, $\Delta_{sol}H > 0$): Solubility **increases** as temperature rises.
  - If dissolution releases heat (*exothermic*, $\Delta_{sol}H < 0$): Solubility **decreases** as temperature rises.
- **Effect of Pressure:** Pressure has almost no effect because solids and liquids are highly incompressible.

B. Solubility of a Gas in a Liquid

Unlike solids, the solubility of gases is greatly affected by pressure and temperature.

- **Effect of Pressure (Henry's Law):** At a constant temperature, the solubility of a gas is directly proportional to its partial pressure above the liquid.
  Formula: $p = K_H \times x$
  Where p is the partial pressure, x is the mole fraction of the gas in solution, and K_H is Henry's Law constant.
  - Higher K_H means lower gas solubility.
- **Effect of Temperature:** Solubility of gases **decreases** with a rise in temperature. Dissolution of gas acts like condensation (exothermic). Because K_H increases with temperature, solubility drops. This is why aquatic life prefers cold water (it holds more dissolved oxygen).
- **Applications of Henry's Law:**
  - Soft drink bottles are sealed under high pressure to increase CO_2 solubility.
  - Scuba divers breathe gas mixtures diluted with Helium to prevent toxic concentrations of dissolved nitrogen and painful "bends" when surfacing.
  - At high altitudes, low oxygen pressure leads to low dissolved oxygen in the blood (Anoxia).

---

4. Vapour Pressure of Liquid Solutions

When volatile liquids are kept in a closed container, they evaporate and establish an equilibrium between the liquid and vapour phases.

A. Liquid-Liquid Solutions (Raoult's Law)

Raoult’ s Law states that for a solution of volatile liquids, the partial vapour pressure of each component is directly proportional to its mole fraction in the solution.
$p_1 = p_1^0 \times x_1$ and $p_2 = p_2^0 \times x_2$
(Where p^0 is the vapour pressure of the pure liquid component).

- **Total Vapour Pressure (Dalton's Law):** The total pressure over the solution is the sum of the partial pressures: $p_{total} = p_1 + p_2$.
  This shows that total vapour pressure varies linearly with the mole fraction of the components.
- **Vapour Phase Composition:** The mole fraction of a component in the *vapour phase* (y_1) can be found using: $y_1 = \frac{p_1}{p_{total}}$.
- **Special Case:** Raoult's law is a special case of Henry's Law where the proportionality constant K_H becomes exactly equal to p^0.

B. Solid-Liquid Solutions (Non-Volatile Solute)

When a **non-volatile solute** (like salt or sugar) is added to a volatile solvent (like water), the vapour pressure of the solution is **lower** than that of the pure solvent.

- **Reason:** Solute molecules occupy surface area, reducing the fraction of surface covered by solvent molecules. Fewer solvent molecules escape into the vapour phase, reducing vapour pressure.

---

5. Ideal and Non-Ideal Solutions

Liquid-liquid solutions are classified into two categories based on how well they obey Raoult's law.

A. Ideal Solutions

Ideal solutions obey Raoult's Law over the *entire range of concentration*.

- **Thermodynamic Properties:**
  - Enthalpy of mixing is zero ($\Delta_{mix}H = 0$): No heat is absorbed or evolved.
  - Volume of mixing is zero ($\Delta_{mix}V = 0$): Volume is perfectly additive.
- **Molecular Reason:** The intermolecular attractive forces between the mixed components (A-B) are nearly equal to the pure components (A-A and B-B).

- **Examples:** n-hexane + n-heptane, benzene + toluene.

B. Non-Ideal Solutions

These solutions do not obey Raoult's law and show deviations (their vapour pressure is higher or lower than predicted).

- **Positive Deviation:**
  - Vapour pressure is **higher** than expected.
  - A-B interactions are **weaker** than A-A or B-B interactions, making it easier for molecules to escape into vapour.
  - Example: Ethanol + Acetone.
- **Negative Deviation:**
  - Vapour pressure is **lower** than expected.
  - A-B interactions are **stronger** than A-A or B-B interactions, decreasing the escaping tendency of molecules.
  - Example: Phenol + Aniline, Chloroform + Acetone.

C. Azeotropes

Binary mixtures formed by highly non-ideal solutions that have the **same composition in both the liquid and vapour phases**. They boil at a constant temperature and cannot be separated by simple fractional distillation.

- **Minimum Boiling Azeotropes:** Formed by solutions exhibiting large *positive* deviations from Raoult's law (e.g., 95% ethanol in water).
- **Maximum Boiling Azeotropes:** Formed by solutions exhibiting large *negative* deviations from Raoult's law (e.g., 68% nitric acid in water).